## INSTALLING THE NEUTRAL COMPONENTS

11. **Refer to Figure 7**
    If branch neutral terminal strips (Item 18) are NOT already installed -- or are full -- then the branch neutral terminal strip (included, in a bag labeled “Branch Neutral Lug Kit”) must be installed. Depending upon the feed location, the branch neutral can be mounted near the bottom of the Z-rails -- or mirrored to the top -- on a branch neutral connector. Panelboards should have one mounted already. On Switchboards, mount the branch neutral connector (Item 17) on the neutral bus using (2) 1/4”-20 screws (Item 15). Mount the neutral terminal strip (Item 18) on either side of the branch neutral connector using (2) #8-32 screws (Item 16). Torque all hardware as follows: For 1/4”-20 screws, refer to torque on label. For #8-32 screws, torque to 35 lb-in.

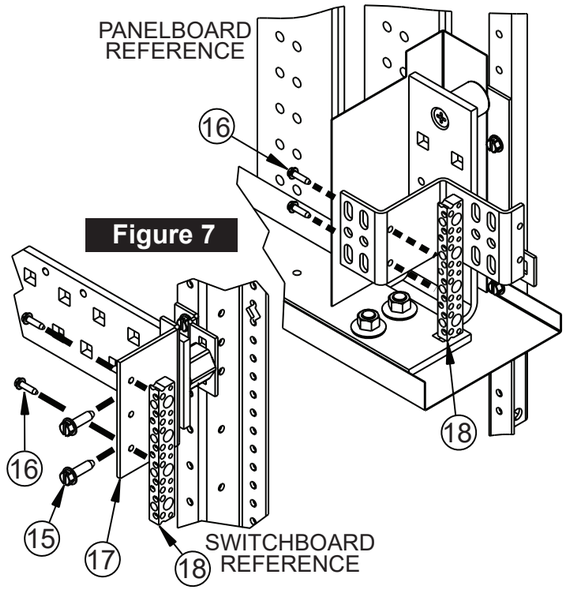



**Figure 7**

## INSTALLING COVER PLATE AND FILLERS

12. Reassemble the deadfront assembly to the interior (removed in Step 3).

13. **Refer to Figure 8**
    Install the 3.75” cover plate (Item 7) by aligning it over the 3.75” of selected unit space and related breakers. Use two 1/4”-20 x 3/8” screw (Item 14), one on each side of the cover plate, and drive into the deadfront rails. Torque all hardware as specified on the rear of the deadfront. Mark the circuit ID cards (Item 8) as desired and mount them to the cover plate by placing each behind a clear card holder (Item 9) and pushing the two into the proper pair of holes in the cover plate next to the proper circuit. Repeat for each location. Finally, use blank fillers (Item 10) at every location that doesn’t have a breaker penetrating through the cover plate.
    NOTE: Other cover plates may be required for proper fit. See filler plate kits listed on the rear of the deadfront.

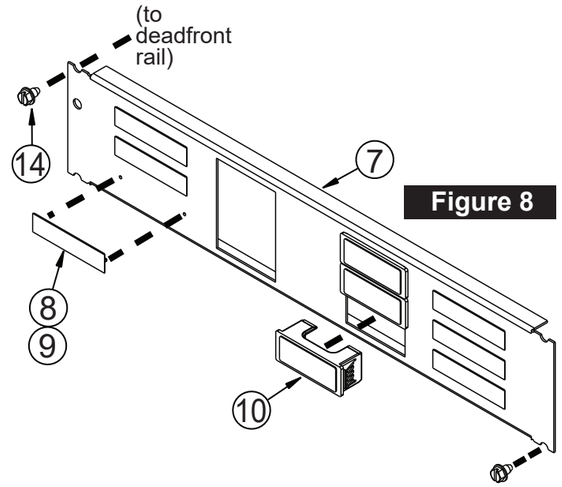



**Figure 8**

## COMPLETING THE INSTALLATION

14. **Ensure that any bus or breaker hardware is torqued as specified on the rear of the deadfront. Then replace all the cover plates and gutter covers. Finally, ensure any fillers are in place BEFORE re-energizing the equipment.**

These instructions do not purport to cover all details or variations in equipment, or to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise, which are not covered sufficiently for the purchaser’s purposes, the matter should be referred to the local **Siemens** sales office. The contents of this instruction manual shall not become part of or modify any prior or existing agreement, commitment or relationship. The sales contract contains the entire obligation of **Siemens**. The warranty contained in the contract between the parties is the sole warranty of **Siemens**. Any statements contained herein do not create new warranties or modify the existing warranty.

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SIEMENS

# S3VA41T / S3VA41TD

## P4 and P5 Power Panel / Distribution Switchboard Kit for Branch Mounting of Type 3VA41 or xGB Breakers

**Installation Instructions**

11-B-1057-01 Rev. 00

# Installation Instructions

The following instructions are for the installation of **Siemens Type 3VA41 (SEAB, MEAB or HEAB)** or **Type xGB** (**NGB**, **HGB** or **LGB**) breakers as branch devices in a Distribution Switchboard or Type P4 or P5 Panelboard. The parts provided in this kit connect **Type 3VA41** or **xGB** breakers to a 1-phase, 2 or 3-wire system or a 3-phase 3 or 4-wire system. Kit **S3VA41T** is for Type P4 (shallow) installations; kit **S3VA41TD** is for Type P5 or Distribution Switchboard (deep) installations. This kit requires 3.75" of unit space. **NOTE: The breakers are <u>NOT</u> included in this kit and must be purchased separately.**



**⚠ DANGER**

**Hazardous voltage. Will cause death or serious injury.**
Keep out.
Qualified personnel only.
Disconnect and lock off all power before working on this equipment.

1. **Lock off power supplying this equipment before working on it.**

2. **Refer to Figure 1**
   Remove the gutter covers and cover plates.

**Figure 1**

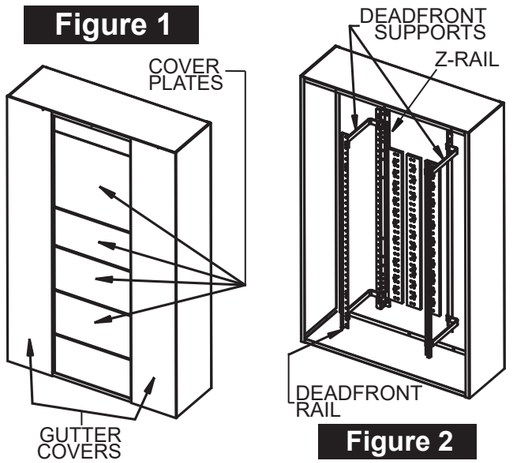



**Figure 2**

3. **Refer to Figure 2**
   To mount a Siemens Type 3VA41 or xGB circuit breaker, 3.75" of empty unit space is required. To locate the proper mounting position, measure from the top deadfront support to the intended location of the empty unit space (filler) on the deadfront. Transfer this dimension from the deadfront support along the Z-rail and mark. This will be the unit space (as shown in Figures 3 & 4). After marking the Z-rail, remove the deadfront rails.

4. If an existing branch module occupies the same location chosen for this kit, remove all of its devices, components and parts.

5. Open the shipping box and inspect the contents.
   **THIS KIT CONTAINS THE FOLLOWING ITEMS:**

| ITEM | DESCRIPTION | QTY |
|---|---|---|
| 1 | A/C-phase Runback * | 2 |
| 2 | A/C-phase Strap ** | 3 |
| 3 | B-phase Runback * | 1 |
| 4 | B-phase Strap ** | 1 |
| 5 | Barrier Support Bracket * | 2 |
| 6 | Top Barrier | 1 |
| 7 | 3.75" Cover Plate | 1 |
| 8 | Circuit ID Card | 6 |
| 9 | Card Holder | 6 |
| 10 | Blank Filler ** | 6 |
| 11 | Bus Hardware Kit, Strap to Section *** | 1 |
| 12 | #10-24 x 3/8" Hex Head Washer Screw | 2 |
| 13 | #10-24 x 5/8" Socket Head Cap Screw | 6 |
| 14 | 1/4"-20 x 3/8" Hex Head Washer Screw | 4 |
| 15 | 1/4"-20 x 5/8" Hex Head Washer Screw ** | 5 |
| 16 | #8-32 x 5/8" Hex Head Washer Screw | 2 |
| 17 | Branch Neutral Connector **** | 1 |
| 18 | Branch Neutral Terminal Strip | 1 |

* Kit contains shorter units for shallow applications (P4) or taller units for deeper applications (P5 or Switchboard).
** Application dictates amount of these components used.
*** This sub-kit of parts is packaged in a bag labeled "Standard Bus Hardware Kit". It includes an instruction card and Qty. 18 of fasteners (three different lengths). Select the five required per the instruction card and discard the rest.
**** Used with Distribution Switchboard (S3VA41TD) applications only.

## INSTALLING THE MOUNTING BRACKETS

6. **Refer to Figure 3**
   Locate a 0.228" diameter hole in the left Z-rail 1.88" below the top of the empty unit space selected. Insert a 1/4"-20 x 3/8" screw (Item 14) through the barrier support bracket (Item 5) mounting hole and into the selected hole in the Z-rail (orienting the bracket as shown). Repeat this for the second mounting bracket onto the right-side Z-rail. Tighten these two screws (Item 14).

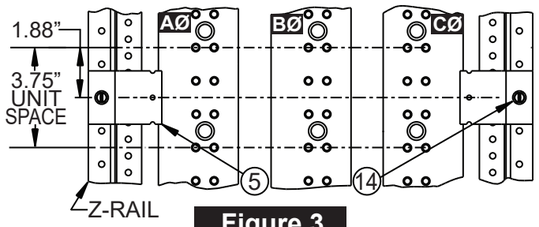



**Figure 3**



**<u>Options for installation of runbacks and straps</u>**
Both kit **S3VA41T** (for Type P4) and kit **S3VA41TD** (for Type P5 or Distribution Switchboard) have components for either single-phase OR three-phase systems. Follow the instructions on this page for your specific system.

## INSTALLING THE RUNBACKS

7. **Refer to Figure 4**
   Locate (2) 0.228" diameter holes in the A-phase bus 1.25" down from the top of the 3.75" empty unit space selected. Insert (2) 1/4"-20 screws (see mounting hardware kit instructions located in Item 11 for the proper length) through the two mounting holes in the A/C runback (Item 1) and into the (2) selected holes in the A-phase bus (orienting the A/C strap as shown) and tighten. Repeat this step to mount the second runback to the C-phase bus.
   **<u>For three-phase installation ONLY:</u>**
   (Skip this for single-phase systems)
   Insert a 1/4"-20 screw (Item 11) through the mounting hole in the B-phase runback (Item 3) and into the proper 0.228" hole located in the B-phase bus, 1.25" up from the bottom of the 3.75" empty unit space (orienting the B-phase runback as shown) and tighten.
   **<u>For either installation:</u>** Torque the 1/4"-20 screws as specified on the rear of the deadfront.

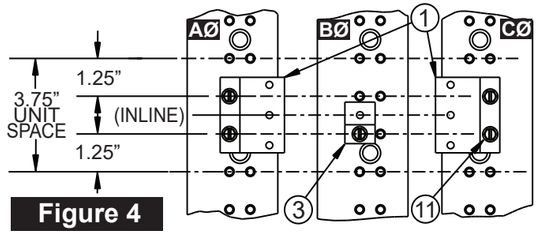


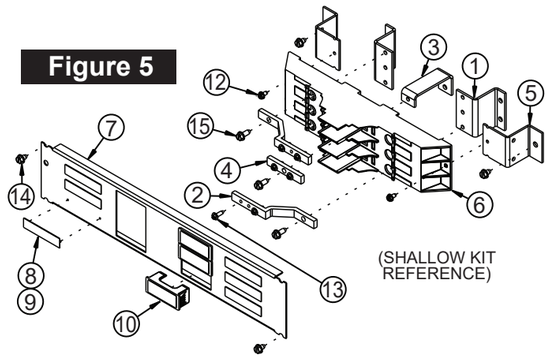

**Figure 4**

**(This figure shows three-phase installation.)**

## INSTALLING THE BARRIER

8. **Refer to Figure 5**
   Set the top barrier (Item 6) over the two barrier support brackets (Item 5) such that the barrier mounting holes align with the holes on the brackets. Insert a #10-24 x 3/8" screw (Item 12) on each side of the barrier and tighten.

**Figure 5**



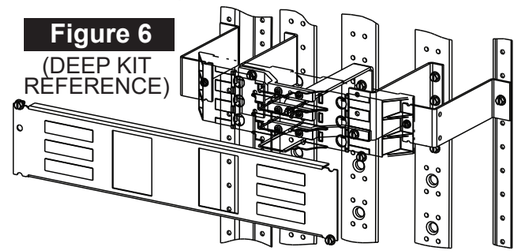

## INSTALLING THE STRAPS

9. **Refer to Figure 5**
   **<u>For three-phase installation ONLY:</u>**
   (Skip this for single-phase systems)
   To install the B-phase strap (Item 4), locate it in the center opening such that the center hole lines up with the mounting hole on the B-phase runback. Drive a 1/4"-20 x 5/8" screw (Item 15) thru the strap and into the runback and tighten. For the A & C-phase straps (Item 2) take one and angle the mounting side through the corresponding hole in the barrier such that the strap mounting hole aligns with the appropriate hole in the A-phase runback. Drive a 1/4"-20 x 5/8" screw (Item 15) through the strap and into the runback and tighten. Similar, but opposite, for the C-phase strap.
   **<u>For single-phase installation:</u>**
   (Skip this for three-phase systems)
   To install only A & C-phase straps, alternate the straps: Either A-C-A or C-A-C pattern. Take an A & C-phase strap (Item 2) and angle the mounting side through the corresponding hole in the barrier such that the strap mounting hole aligns with the appropriate hole in the A or C phase runback. Drive a 1/4"-20 x 5/8" screw (Item 15) through the strap and into the runback and tighten. Similar for the other straps, alternating assembly to the A and C-phase bus.
   **<u>For either installation:</u>** Torque the 1/4"-20 screws as specified on the rear of the deadfront.

## INSTALLING THE SIEMENS BREAKER

10. **Refer to Figure 6 and Figure 6A**
    Orient the breaker so that the breaker stabs are to the inside, and align the bottom of the Type 3VA41 or xGB breaker with the top of the barrier. Slide the breaker along until the notch on the bottom of the breaker fully engages the edge of the barrier. Once fully engaged, use #10-24 x 5/8" screws (Item 13) to secure each of the breaker pole stabs to the properly aligned straps. Torque all hardware as specified on the rear of the deadfront.

**Figure 6**



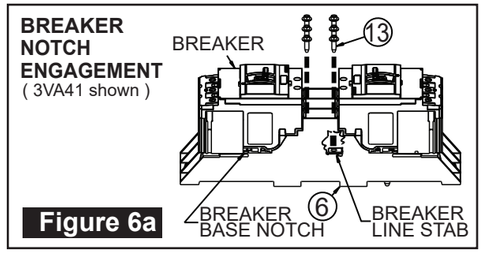




**Figure 6a**

**(more instructions on following page)**



11-B-1057-01 Rev. 00